

5 The Hawaiian islands have 18 species of honeycreeper birds.

The Hawaii amakihi honeycreeper and the Hawaii palila honeycreeper are two species found on these islands.

(a) The table gives information about these two species of honeycreeper.

Information	Hawaii amakihi honeycreeper	Hawaii palila honeycreeper
Photograph	 (Source: © NPS Photo/Alamy Stock Photo)	 (Source: © Minden Pictures/Alamy Stock Photo)
Habitat	lowland and mountainous forest and shrubland	mountainous forest
Number of Hawaiian islands with a population	3	1
Estimated population size	<10 000	<1000
Food source	nectar, tree sap, spiders and insects	seeds and berries

- (i) The Hawaii palila honeycreeper is found only in the mountainous forest on one island.

Which of the following is the term used to describe this distribution?

(1)

- ☐ **A** diversity
- ☐ **B** endemic
- ☐ **C** polygenic
- ☐ **D** specific

- (ii) In 2003, it was estimated that there were 7100 Hawaii palila honeycreeper birds on this island. In 2020, it was estimated that there were 950 of these birds.

Calculate the percentage change in the estimated populations of these birds.

Give your answer to **two** significant figures.

(2)

Answer %



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- (c) Explain how scientists could determine which of the 17 other species of honeycreeper on these islands is the most closely related to the Hawaii palila honeycreeper species.

(2)

(Total for Question 5 = 11 marks)

